



Esophageal and Gastric Tumors

Carcinoma of the Esophagus

Oesophageal cancer is relatively rare (4: 100 000) population Common in Iran, parts of Africa and China In Iraq, incidence rate of esophageal tumor was 0.57/100 000 populations in 2018. Despite modern treatment, the overall 5-year survival of patients presenting with oesophageal cancer is only 13%.

Histological subtypes:

- Squamous oesophageal cancer
- Adenocarcinomas of oesophagus

Squamous oesophageal cancer can occur in any part of the oesophagus, and almost all tumours in the upper oesophagus are squamous cancers.

Etiology of the esophageal carcinoma

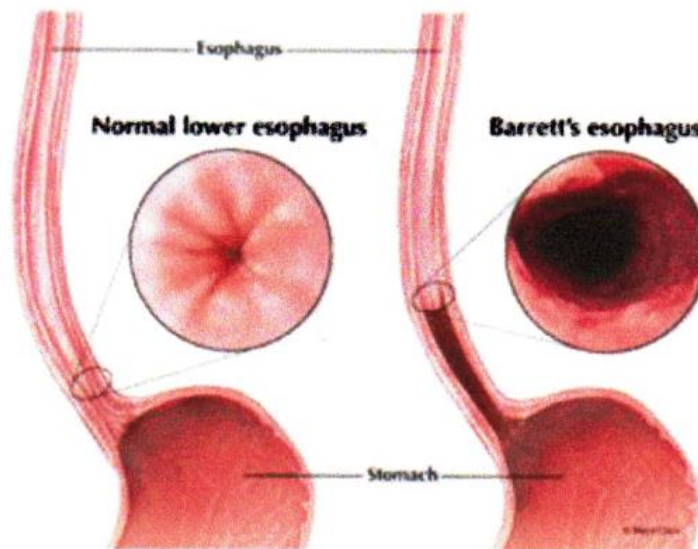
- Smoking
- Alcohol excess
- Chewing tobacco
- Achalasia of the esophagus
- Coeliac disease
- Stricture
- Post-caustic stricture
- Tylosis (familial hyperkeratosis of palms and soles)



Adenocarcinomas typically arise in the lower third of the oesophagus from Barrett's oesophagus or from the cardia of the stomach.

The incidence is increasing in Western country this is possibly because of the high prevalence of gastro-oesophageal reflux and Barrett's oesophagus in Western populations.

High body mass index is regarded as risk factor for adenocarcinoma.



Clinical features

Symptoms:

- Dysphasia: progressive, painless for solid foods
- Weight loss, chest pain, hoarseness of voice, suggesting mediastinal invasion
- Coughing after swallowing may be due to fistulation
- Pneumonia and pleural effusion

Signs:

May be absent, cachexia, cervical lymphadenopathy or other evidence of metastatic spread is common.



Investigation

1- Upper gastrointestinal endoscopy with biopsy

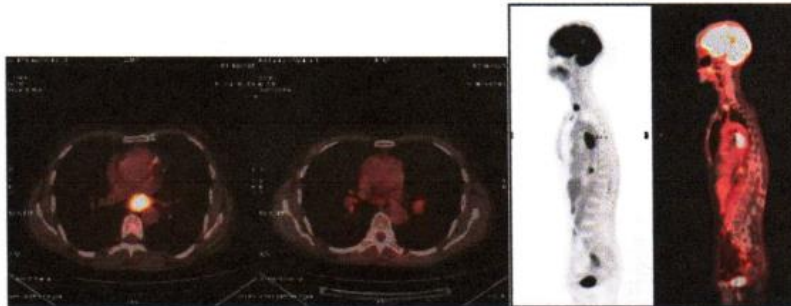
2- Barium swallow

After Diagnosis, for staging and defining operability;

3- CT of chest and abdomen

4- PET-CT

5- Endoscopic U/S to determine the depth of penetration of the tumour into the oesophageal wall and to detect locoregional lymph node involvement.



Positron emission tomography-computed tomography (PET-CT) staging of oesophageal carcinoma. Whole-body PET scan showing avid uptake in the primary tumour but also in distant paratracheal (superior thin arrow) and gastro-oesophageal lymph nodes

Esophageal tumor





Staging

Table 1 T, N, and M status and histologic grade definitions for esophagus and esophagogastric junction cancer in the 7th edition of the American Joint Committee on Cancer (AJCC) Cancer Staging Manual

T status

T _{is}	High-grade dysplasia
T1	Invasion into the lamina propria, muscularis mucosae, or submucosa
T2	Invasion into muscularis propria
T3	Invasion into adventitia
T4a	Invades resectable adjacent structures (pleura, pericardium, diaphragm)
T4b	Invades unresectable adjacent structures (aorta, vertebral body, trachea)

N status

N0	No regional lymph node metastases
N1	1 to 2 positive regional lymph nodes
N2	3 to 6 positive regional lymph nodes
N3	7 or more positive regional lymph nodes

M status

M0	No distant metastases
M1	Distant metastases

Histologic grade

G1	Well differentiated
G2	Moderately differentiated
G3	Poorly differentiated
G4	Undifferentiated

Staging of ET

Table 2 AJCC 7th edition stage groupings

Stage	Adenocarcinoma				Squamous cell carcinoma				
	T	N	M	Grade	T	N	M	G	Location
0	is	0	0	1	is	0	0	1	Any
IA	1	0	0	1-2	1	0	0	1	Any
IB	1	0	0	3	1	0	0	2-3	Any
	2	0	0	1-2	2-3	0	0	1	Lower
IIA	2	0	0	3	2-3	0	0	1	Upper, middle
					2-3	0	0	2-3	Lower
IIB	3	0	0	Any	2-3	0	0	2-3	Upper, middle
	1-2	1	0	Any	1-2	1	0	Any	Any
IIIA	1-2	2	0	Any	1-2	2	0	Any	Any
	3	1	0	Any	3	1	0	Any	Any
	4a	0	0	Any	4a	0	0	Any	Any
IIIB	3	2	0	Any	3	2	0	Any	Any
IIIC	4a	1-2	0	Any	4a	1-2	0	Any	Any
	4b	Any	0	Any	4b	Any	0	Any	Any
	Any	3	0	Any	Any	3	0	Any	Any
IV	Any	Any	1	Any	Any	Any	1	Any	Any

Cancer location definitions: upper thoracic, 20-25 cm from incisors; middle thoracic, 25-30 cm from incisors; lower thoracic, 30-40 cm from incisors.



Management

Could be radical or palliative;

- **Surgery** if the tumour is confined to the oesophageal wall and there is no spread to lymph nodes. Invasion of the aorta, major airways or coeliac axis usually precludes surgery. For very early superficial tumours, endoscopic submucosal dissection may offer an alternative to surgery but is not widely used outside of Japan and Korea.
- **Neoadjuvant** chemotherapy for large tumor.
- **Chemoradiation** increase survival in unresectable locally advanced disease.
- **Palliative for Metastatic disease** should focus on relief of dysphagia and pain. Endoscopic laser therapy or self-expanding metallic stents can be used to improve swallowing. Palliative radiotherapy may induce shrinkage of tumor.
- Nutritional support and appropriate analgesia may improve the quality of life.

Prognosis

Survival rates depend on several factors, including the stage of the cancer when it was first diagnosed, location of the tumor and performance status of the patient.

The 5-year survival rate for stage, II is around 47%.

The 5-year survival rate for stage III is around 25%.

If it has spread to distant parts of the body, the survival rate is less than 5%.

Approximately 70% of patients have extensive disease at presentation, treatment is palliative and should focus on relief of dysphagia and pain.



Gastric Carcinoma

Epidemiology

Gastric adenocarcinoma is the third leading cause of cancer related death worldwide.

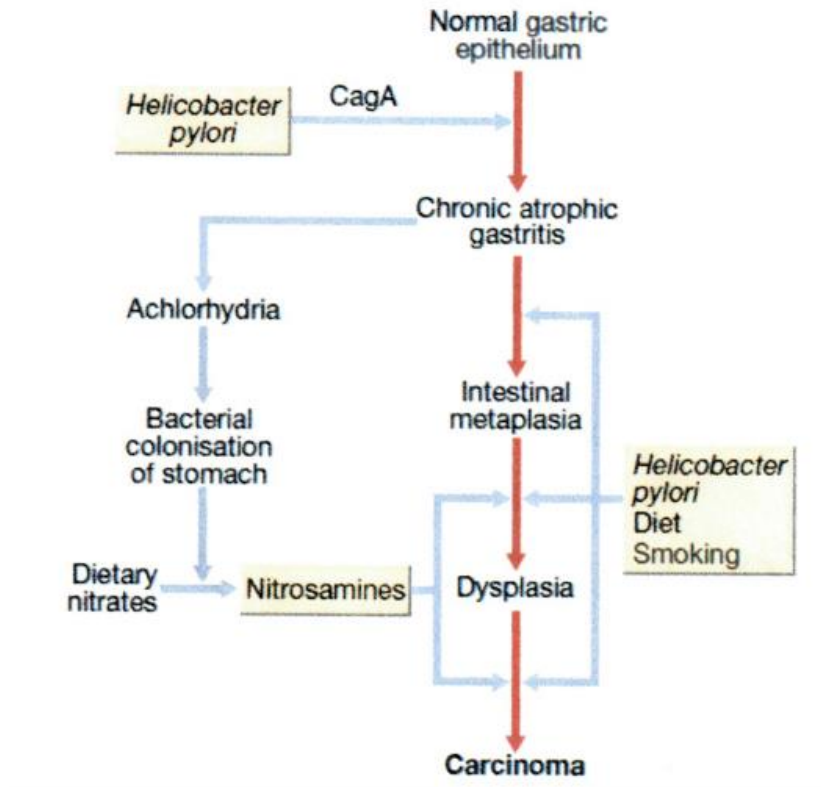
There is marked geographical variation in incidence, more common in China, Japan, Korea than Eastern Europe and parts of South America. Japanese migrants to the USA have revealed a much lower incidence in second generation migrants.

The incidence is 50% lower in women. In both sexes, it rises sharply after 50 years of age.

In Iraq, incidence rate of Gastric tumor was 2.57/100 000 populations in 2018.

Etiology: Multifactorial

- *H. pylori* (contribute in 60-70%)
- Smoking
- Alcohol
- Dietary factors (salted, smoked foods and the consumption of nitrates, lacking fresh fruit and vegetables, as well as vitamins C and A)
- Autoimmune gastritis (pernicious anaemia)
- Adenomatous gastric polyps
- Previous partial gastrectomy (>20 yrs)
- Links with blood group A
- Ménétrier's disease
- Familial adenomatous polyposis



Infection with *H. pylori* plays a key pathogenic role and the infection has been classified by the International Agency for Research on Cancer (IARC) as a definite human carcinogen. It is associated with chronic atrophic gastritis, gastric mucosal atrophy and gastric cancer. It has been estimated that *H. pylori* infection may contribute to the occurrence of gastric cancer in 70% of cases.

Although the majority of *H. pylori*-infected individuals have normal or increased acid secretion, a few become hypo- or achlorhydric and these people are thought to be at greatest risk. *H. pylori*-induced chronic inflammation with generation of reactive oxygen species and depletion of the normally abundant antioxidant ascorbic acid are also important.

Several strain-specific virulence factors were reported, One of the main virulence factors is CagA, which is associated with higher risk of gastric cancer and peptic ulcer. CagA protein can interact with intercellular proteins and activate signaling pathways through both tyrosine the infection causes an antral-predominant pattern of gastritis characterised by hypergastrinaemia and a very exaggerated acid production by parietal cells, which could lead to duodenal ulceration. In a much smaller number of infected people, *H. pylori* causes



a corpus-predominant pattern of gastritis leading to gastric atrophy and hypochlorhydria. This phenotype is much more common in Asian countries ,particularly Japan, China and Korea. The hypochlorhydria allows other bacteria to proliferate within the stomach; these other bacteria continue to drive the chronic inflammation and produce mutagenic nitrites from dietary nitrates, predisposing to the development of gastric cancer consumption of nitrites and nitrates may increase cancer risk.

Carcinogenic N-nitroso-compounds are formed from nitrates by the action of nitrite-reducing bacteria that colonise in the achlorhydric stomach.

No predominant genetic abnormality has been identified, although cancer risk is increased two- to threefold in first-degree relatives of patients, and links with blood group A have been reported. Some host genetic factors related to inflammatory genes have recently been associated with increased risk of gastric cancer.

Pathology

Adenocarcinomas arising from mucus-secreting cells in the base of the gastric crypts. develop on a background of chronic atrophic gastritis with intestinal metaplasia and dysplasia.

Cancers are either:

- 1- Intestinal, arising from areas of intestinal metaplasia with histological features reminiscent of intestinal epithelium.
- 2- Diffuse, arising from normal gastric mucosa, tend to be poorly differentiated and occur in younger.

Anatomical distribution

50% develop in the antrum.

20-30% in the gastric body, (greater curve).

20% are found in the cardia.



In developing countries, distal tumor is more common than proximal, however in Western countries, proximal tumor is more commoner, this change in disease pattern may be a reflection of changes in lifestyle or the decreasing prevalence of *H. pylori* in the West.

Gross appearances

- 1- Ulcerative
- 2- Polypoidal or fungating
- 3- Diffuse submucosal infiltration by a scirrhus cancer (linitis plastica) is uncommon

Early gastric cancer is defined as cancer confined to the mucosa or submucosa



TNM staging of GT

Primary Tumor

- Tis: Carcinoma *in situ*
- T1: Invasion of lamina propria or submucosa
- T2: Invasion of muscularis propria
- T3: Invasion of serosa
- T4: Invasion of adjacent structures

Lymph Node Status

- N0: No regional lymph node involvement
- N1: Metastases to 1 to 2 regional lymph nodes
- N2: Metastases to 3 to 6 regional lymph nodes
- N3: Metastases to more than 7 regional lymph nodes

Metastatic Disease

- M0: No distant metastases
- M1: Distant metastases present



stages	T	N	M
I A	T1	N0	M0
I B	T1	N1	M0
	T2	N0	M0
II	T1	N2	M0
	T2	N1	M0
	T3	N0	M0
III	T2	N2, N3	M0
	T3	N1, N2, N3	M0
	T4	ANY N	M0
IV	Any T	Any N	M1

Clinical features

Early gastric cancer is usually asymptomatic but may be discovered during screening processes. in Japan 2/3 of patients with advanced cancers.

Symptoms:

Weight loss and 50% have ulcer-like pain. Anorexia and nausea, early satiety, haematemesis, melaena and dyspepsia. Dysphagia occurs in tumours of the gastric cardia that obstruct the gastro-oesophageal junction. Anaemia from occult bleeding is also common.

Suspicious clinical features: patients with dyspepsia and alarm features which include:

- Weight loss
- Anaemia
- Haematemesis, melaena
- Dysphagia
- Palpable abdominal mass

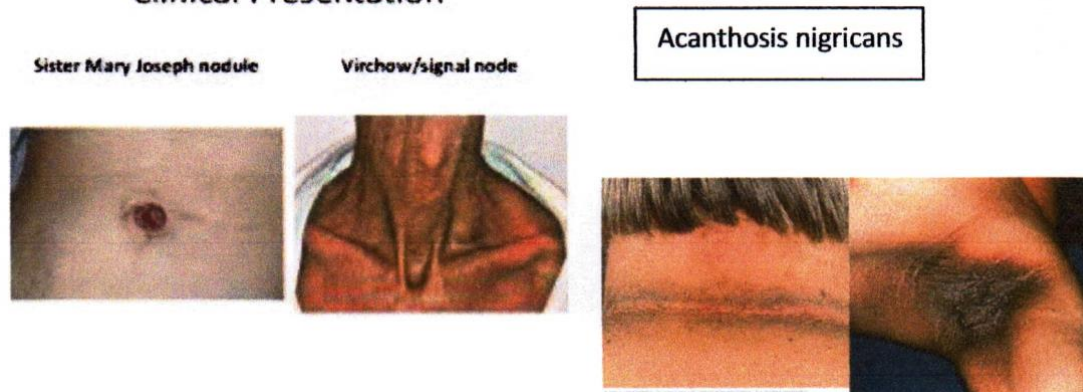
**Signs:**

Signs of weight loss, anaemia and a palpable epigastric mass are not infrequent.

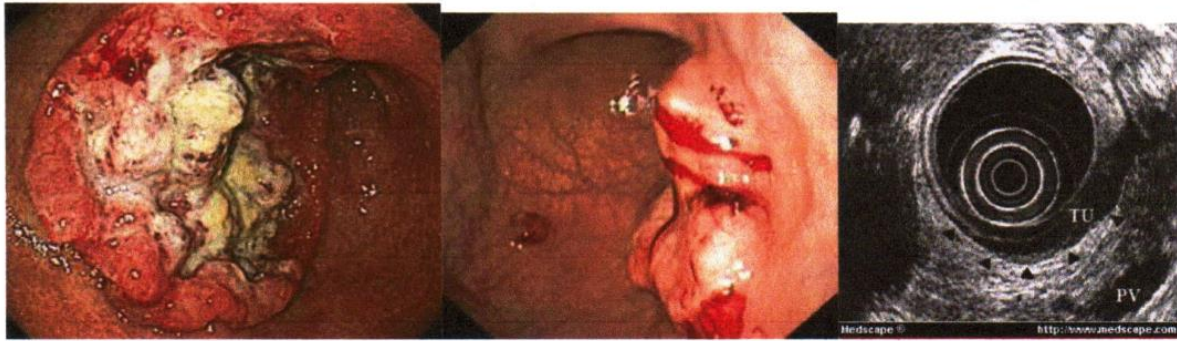
Jaundice or ascites signify metastatic spread, supraclavicular lymph nodes(Troisier's sign), umbilicus (Sister Joseph's nodule) or ovaries (Krukenberg tumour).

- Paraneoplastic phenomena, such as acanthosis nigricans, thrombophlebitis and dermatomyositis, occur rarely.

Metastases arise most commonly in the liver, lungs, peritoneum and bone marrow.

Clinical Presentation**Investigations**

- 1- Upper gastrointestinal endoscopy (OGD): highly sensitive and specific diagnostic test. multiple biopsy specimens should be obtained from any visually suspicious areas.
- 2- Barium meal is a poor alternative test.
- 3- Further imaging is necessary for staging and assessment of resectability.
 - U/S of abdomen
 - CT of abdomen
 - PET-CT scan
 - Endoscopic U/S
 - Laparoscopy with peritoneal washings



Management

Surgery

Resection offers the only hope of cure, and this can be achieved in about 90% of patients with early gastric cancer.

For the majority of patients with locally advanced disease, total gastrectomy with lymphadenectomy. Proximal tumours involving the oesophago-gastric junction also require a distal oesophagectomy.

Palliative resection may be necessary when patients present with bleeding or gastric outflow obstruction.

Perioperative chemotherapy improves survival rates. (Epirubicin, Cisplatin, 5FU)

Palliative treatment

In patients with inoperable tumours, survival can be improved and palliation of symptoms achieved with chemotherapy.

Metastatic gastric cancer has a poor outlook, with survival around 4 months with best supportive care. In patients with inoperable tumours, survival can be improved to 12 months and palliation of symptoms achieved with chemotherapy.

The biological agent trastuzumab may benefit some patients whose tumours over-expressed HER2.

Endoscopic laser ablation for control of dysphagia or recurrent bleeding.



Endoscopic dilatation or insertion of expandable metallic stents for relief of dysphagia or vomiting.

Nasogastric tube may offer temporary relief of vomiting due to gastric outlet obstruction.

Gastric Lymphoma

This is a rare tumour accounting for less than 5% of all gastric malignancies.

The stomach is the most common site for extranodal non-Hodgkin lymphoma and 60% of all primary gastrointestinal lymphomas occur at this site.

The majority of gastric lymphoma is low grade lymphoma (classified as extranodal marginal zone lymphomas of MALT type), *H. pylori* infection is closely associated with the development. Other type of gastric lymphoma is High grade B cell lymphoma.

The clinical presentation is similar to that of gastric cancer. endoscopically the tumour appears as a polypoid or ulcerating mass.

Management

Depend on grade, extend of tumor and performance status:

- *H. Pylori* eradication for early (superficial) tumor low grade lymphoma.

Combination chemotherapy and Rituximab (anti-CD20) for B cell high grade lymphoma.

- Surgery and radiotherapy may be used.

Other Tumors of Stomach

Gastrointestinal stromal cell tumours (GIST) arising from the interstitial cells, expression of the *c-kit proto-oncogene*, which encodes a tyrosine kinase receptor.

These tumours are usually asymptomatic, but may occasionally be responsible for dyspepsia, ulceration and gastrointestinal bleeding.

**Treatment**

- Surgery
- Imatinib (a tyrosine kinase inhibitor) to reduce their size and make surgery easier. Imatinib can also provide prolonged control of metastatic GISTs.

Hyperplastic polyps and Adenomatous polyps are rare but have malignant potential and should be removed endoscopically.

Gastric carcinoid tumours

These benign tumours arise from enterochromaffin-like (ECL) cells or endocrine cells, and are often multiple but rarely invasive (unlike carcinoid tumor that arise elsewhere from GIT) they usually run a benign and favorable course. However, large (>2 cm) carcinoids may metastasis and should be removed.

لا تنسوا أهلنا في فلسطين و السودان و سوريا من صادق دعائكم